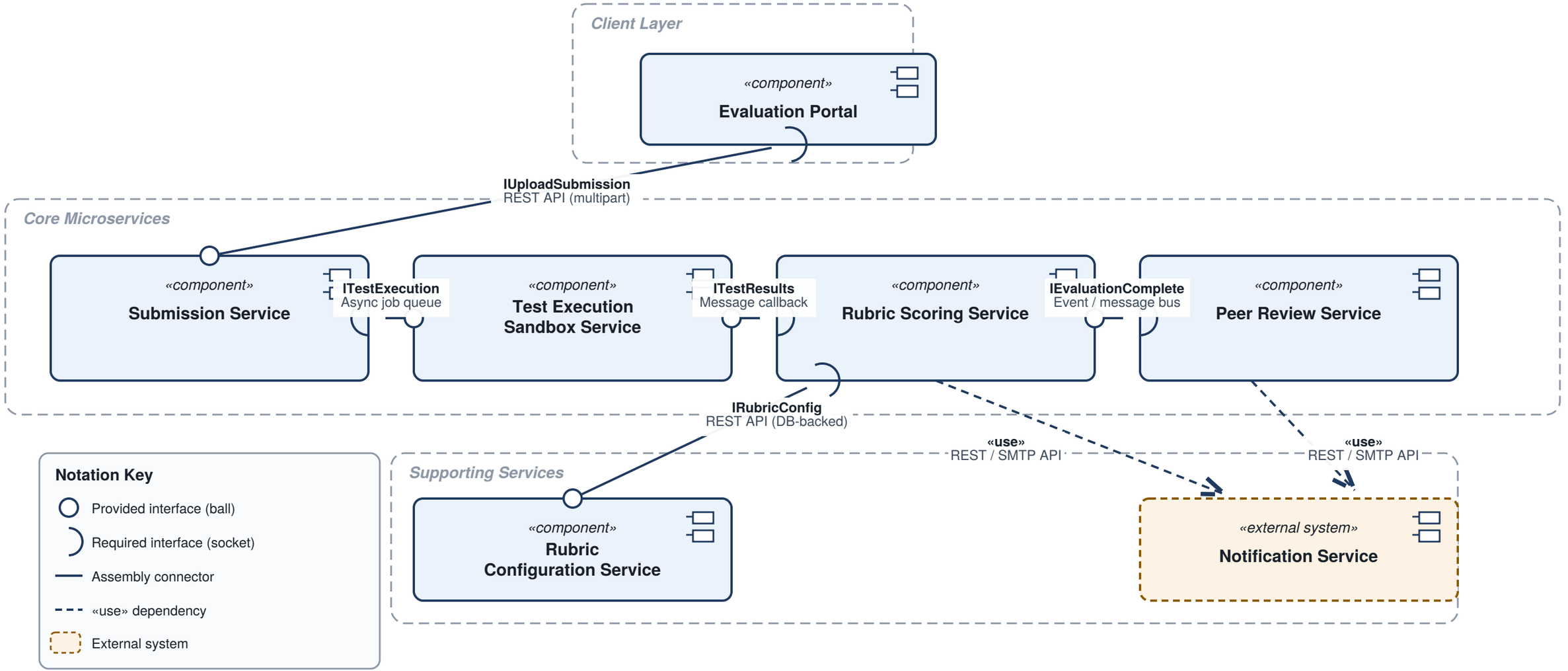
Automated Rubric Assignment Evaluator — UML Component Diagram

Architectural Style: Microservices Architecture



Lab 3: Component Modelling & Architectural Pattern Selection

Written Justification — Automated Rubric Assignment Evaluator

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Section:	E	**Date:**	08/09/26

Architecture Selection

We chose **Microservices Architecture** for the Automated Rubric Assignment Evaluator.

Reason 1 — Independent scaling under bursty concurrent load

NFR-001 requires the system to handle up to 100 concurrent submissions near assignment deadlines without dropping queue items or exceeding a 1.5 GB memory footprint. Splitting the Test Execution Sandbox Service from the Submission, Rubric Scoring, and Peer Review services lets only the sandbox workers be scaled out during a deadline rush, instead of scaling an entire monolith, which matches this bursty, uneven-load profile far better than a Layered or Client-Server design.

Reason 2 — Fault isolation across independently evolving workflows

The system bundles several loosely related workflows — automated evaluation, faculty rubric configuration, peer-review assignment, and notifications — each owned by a different actor (Student, Faculty Evaluator, Peer Reviewer) and each able to change or fail independently. Microservices let the Test Execution Sandbox Service crash or time out (per the alternate flow in the use-case spec) without bringing down Rubric Configuration, Peer Review, or Notification, so one failing submission cannot stall the whole evaluator.

Security Advantage

NFR-002 requires each submission's test suite to run inside an isolated sandbox so submitted code cannot reach other submissions, the host filesystem, or the network. Deploying the Test Execution Sandbox Service as its own microservice, reachable only through the ITestExecution/ITestResults interfaces, means a malicious or buggy student submission is contained inside that one service boundary and cannot directly access the Rubric Configuration Service's data or the Notification Service's credentials.

Performance Benefit

FR-001 requires a rendered rubric breakdown within 10 seconds of submission. Because the Submission Service hands work to the Test Execution Sandbox Service through an asynchronous job queue, the Evaluation Portal stays responsive for other students while sandbox workers process jobs in parallel, and the Rubric Scoring Service can begin scoring as soon as results arrive rather than waiting on a single shared process.